

k: A Theory of Everything

Quantum Field Theory as $\aleph_0$, General Relativity as $\mathfrak{c}$,
and the Kolmogorov Merge of Sampled Universes

k

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Abstract

A theory of everything must say how the two successful infinities of twentieth-century physics communicate. Quantum field theory is the physics of a separable Hilbert space: a countable orthonormal basis, occupation numbers, Fock modes, programs. That is Cantor's infinity of the integers, $\aleph_0$. General relativity is the physics of a smooth Lorentzian 4-manifold: uncountably many points, a metric of cardinality $\mathfrak{c} = 2^{\aleph_0}$. That is Cantor's infinity of the reals. The continuum is the power set of the integers, so a classical universe is a completion of a countable quantum description. We take this as ontology, not metaphor.

Universes are sampled from the quantum state of the universe. Each sample is a finite string—Cauchy data, a code, a program—drawn from the integer skeleton of that state. General relativity inflates the string to a continuum spacetime. The weight of the resulting universe is the gravitational Kolmogorov complexity of the sample: Einstein's equation is the shortest program that extends 3-data to a 4-geometry, and 2^{-K_G} is the corresponding Solomonoff measure. The physical world is the merge of these universes against those weights. The merge is a countable sum, hence well-defined; the continuum appears only after inflation. Classical Einstein gravity is the typical merge of low-complexity samples. Quantum field theory is the residual integer dynamics on the merged background. The gravitational path integral, the Born rule, black-hole entropy, the holographic bound, and the cosmological arrow of time are the same object viewed at different resolutions.

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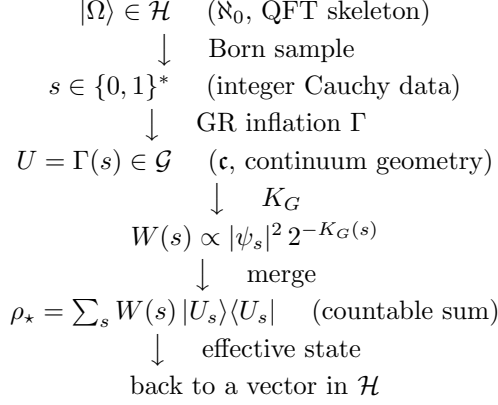


Figure 1: The k -cycle. Quantum field theory supplies a countable state. Sampling reads a finite string from that state. General relativity completes the string to a continuum universe. Gravitational Kolmogorov complexity weights the universe. The merge is a countable sum of those weighted universes and returns an effective quantum state.

1 The problem of two infinities

Physics has two working theories and two working infinities, and they are not the same infinity.

Quantum field theory (QFT) is built on a separable Hilbert space $\mathcal{H}$. Separability means there is a countable orthonormal basis $\{|n\rangle\}_{n \in \mathbb{N}}$. Fock space is assembled from occupation numbers in $\mathbb{N}$. A lattice regularization, a mode expansion, a particle species, a spin component: all of these are indexed by integers. Even $L^2(\mathbb{R}^n)$, whose classical configuration space is a continuum, admits a countable Hilbert basis (Hermite functions, Fourier modes on a box, wavelets). The quantum theory *counts*. Its native cardinal is $\aleph_0$, the infinity of the integers [11, 27].

General relativity (GR) is built on a smooth 4-manifold (M, g) . Any nonempty open 4-manifold has $|M| = \mathfrak{c} = 2^{\aleph_0}$ points. A smooth metric is a section of $\text{Sym}^2 T^*M$; the space of such sections has cardinality $\mathfrak{c}$, because a continuous function is determined by its values on a countable dense set [10]. The diffeomorphism group $\text{Diff}(M)$ likewise has cardinality $\mathfrak{c}$. The Einstein equation is a differential equation on this continuum. GR's native cardinal is $\mathfrak{c}$, the infinity of the reals.

These are not aesthetic preferences. They are the cardinalities of the objects the theories actually use. The obstruction to a theory of everything is not only that one theory is quantum and the other is geometric. It is that one theory lives at $\aleph_0$ and the other lives at $\mathfrak{c}$, and Cantor's theorem says these are different sizes [11]:

$$\aleph_0 < 2^{\aleph_0} = \mathfrak{c}. \quad (1)$$

There is no bijection between the integer skeleton of QFT and the continuum spacetime of GR. Any unification must therefore *supply a map* from one infinity to the other, and a map back.

The map forward is sampling plus inflation. The map back is a weighted merge. The weights are Kolmogorov complexity computed in a language that already contains Einstein's equation. That is the whole theory. The rest of this paper is the mathematics of those three sentences.

2 Cantor, Kolmogorov, Einstein

2.1 Cardinals

Write $|X|$ for the cardinality of a set X . Cantor showed that $|\mathbb{N}| = \aleph_0$ is the smallest infinite cardinal, that $|\mathbb{R}| = \mathfrak{c} = 2^{\aleph_0}$, and that there is no bijection $\mathbb{N} \rightarrow \mathbb{R}$ [9, 11]. The identification $\mathfrak{c} = 2^{\aleph_0}$ is the statement that the continuum is (in bijection with) the power set of the integers: every real in $[0, 1]$ is a subset of $\mathbb{N}$, via its binary expansion

$$r = \sum_{n=1}^{\infty} b_n 2^{-n}, \quad b_n \in \{0, 1\}. \quad (2)$$

The continuum hypothesis CH asserts $\mathfrak{c} = \aleph_1$, i.e. that no cardinal sits strictly between $\aleph_0$ and $\mathfrak{c}$. CH is independent of ZFC [26, 16]. Physics, in this theory, never uses an intermediate cardinal: samples are countable, completions are continuum, and nothing is sampled in between. That is a physical reason to regard CH as true *for physics*, whatever its status in set theory. We return to this in Section 11.

Proposition 2.1 (Cardinalities of the two theories). *Let $\mathcal{H}$ be an infinite-dimensional separable complex Hilbert space, and let (M, g) be a smooth connected Lorentzian 4-manifold. Then:*

1. $\mathcal{H}$ admits a countable orthonormal basis, so the kinematic labels of QFT have cardinality $\aleph_0$.
2. As a set, $|\mathcal{H}| = \mathfrak{c}$.
3. $|M| = \mathfrak{c}$, and the space $C^\infty(M, \text{Sym}^2 T^*M)$ of smooth metrics has cardinality $\mathfrak{c}$.
4. The set of finite binary strings $\{0, 1\}^*$ has cardinality $\aleph_0$.
5. The set of prefix-free programs for a fixed universal Turing machine has cardinality $\aleph_0$.

Proof. (1) is the definition of separability. (2) follows because every vector is an ℓ^2 sequence of complex amplitudes and $|\mathbb{C}^{\mathbb{N}}| = \mathfrak{c}^{\aleph_0} = (2^{\aleph_0})^{\aleph_0} = 2^{\aleph_0 \cdot \aleph_0} = 2^{\aleph_0}$. (3): a nonempty open subset of $\mathbb{R}^4$ has cardinality $\mathfrak{c}$; a continuous real function on a second-countable manifold is determined by its restriction to a countable dense subset, so $|C^0(M, \mathbb{R})| \leq |\mathbb{R}^{\mathbb{N}}| = \mathfrak{c}^{\aleph_0} = \mathfrak{c}$, and the smooth metrics inject into this set. (4) and (5) are countable unions of finite sets. $\square$

The distinction in (1) versus (2) is the entire subject. QFT *uses* a countable basis. Its states, as points of $\mathcal{H}$, already have cardinality $\mathfrak{c}$ —the same cardinal as GR geometries. What QFT does *not* have, and GR does, is a continuum of *independent classical points*. A quantum state is a square-summable sequence. A spacetime is an uncountable set equipped with a smooth metric. The map from the first to the second is not given by the Hilbert space structure alone. It has to be put in by hand as a physical law. That law is sampling plus Einstein inflation.

2.2 Kolmogorov complexity

Let U be a prefix-free universal Turing machine. The prefix Kolmogorov complexity of a finite string $s \in \{0, 1\}^*$ is [34, 13, 45, 36]

$$K(s) = \min\{|p| : U(p) = s\}. \quad (3)$$

$K(s)$ is the length of the shortest program that prints s and halts. It is uncomputable, invariant up to an additive constant under change of $\mathbf{U}$ (the invariance theorem), and it induces the Solomonoff–Levin semimeasure [45, 35]

$$\mathbf{M}(s) = \sum_{\mathbf{U}(p)=s} 2^{-|p|} \geq 2^{-K(s)}. \quad (4)$$

Kraft’s inequality says $\sum_p 2^{-|p|} \leq 1$ for a prefix-free set of programs, so $\mathbf{M}$ is a semimeasure on strings: a universal Occam prior. Schmidhuber proposed that a theory of everything *is* a choice of such a prior on universe-histories [44]. Dzhunushaliev proposed that the probability of a gravitational configuration is $e^{-K[g]}$, and that the Einstein equation is the algorithm that reconstructs a 4-geometry from initial data, thereby collapsing the complexity of Schwarzschild spacetime to the complexity of a single horizon radius plus the complexity of Einstein’s equation [17, 18].

Those two observations are the seed of the weight in this theory. What they do not yet supply is the cardinal split, the sampling map from a quantum state, or the merge.

2.3 Einstein as a compressor

The Cauchy problem of vacuum GR is locally well-posed [15, 31]: a pair (γ, K) of Riemannian 3-metric and extrinsic curvature on a 3-manifold Σ , satisfying the Hamiltonian and momentum constraints, determines a unique (up to diffeomorphism) maximal globally hyperbolic development (M, g) . If the data are analytic, they are fixed by a countable list of Taylor coefficients on a dense set. If the data are specified only to finite precision—and every sample is—they are a finite string.

Remark 2.2 (GR is a compressor). A generic smooth metric on a compact region, written to precision ε in Planck units, requires a number of bits that scales with the 4-volume. An Einstein solution on the same region is determined by 3-data on a Cauchy slice, plus the Einstein equation. The Einstein equation is a short program. Therefore Einstein solutions are exponentially more compressible than generic metrics. This is not an analogy with Kolmogorov complexity. It *is* Kolmogorov complexity, relative to a machine that may call an Einstein solver.

Dzhunushaliev’s estimate for Schwarzschild is the prototype [17]: the whole 4-geometry is recovered from r_g/r_{Pl} together with Einstein’s equation, so

$$K(\text{Schwarzschild}) \approx L\left[\left(\frac{r_g}{r_{\text{Pl}}}\right)^2\right] + L_{\text{Einstein}}, \quad (5)$$

where $L[\cdot]$ is program length for the indicated datum. The Bekenstein–Hawking entropy $A/4\ell_P^2$ is the natural information content of horizon data [3, 32]. We will take this seriously as a theorem schema, not a coincidence (Section 9).

3 Postulates

Five postulates. Nothing else is put in.

Postulate 1 (Integer quantum). The kinematical quantum state of the universe is a vector $|\Omega\rangle$ in a separable Hilbert space $\mathcal{H}$. There exists a countable orthonormal basis $\{|e_n\rangle\}_{n \in \mathbb{N}}$ of $\mathcal{H}$, the *integer skeleton*. All quantum observables used in sampling are defined with respect to this skeleton or a uniformly equivalent one.

Postulate 2 (Continuum gravity). A classical universe is a diffeomorphism class of globally hyperbolic Lorentzian 4-manifolds with matter fields, (M, g, Φ) . Write $\mathcal{G}$ for the space of such classes. $\mathcal{G}$ has cardinality $\mathfrak{c}$. The dynamics on $\mathcal{G}$ is Einstein's equation coupled to the matter whose quantum version lives in $\mathcal{H}$.

Postulate 3 (Sampling). A sample is a finite binary string $s \in \{0, 1\}^*$ obtained by a finite-resolution measurement of $|\Omega\rangle$ on the integer skeleton. The set of samples is countable. Each sample s is interpreted as Cauchy data (geometry plus matter) to finite precision. The *inflation map*

$$\Gamma : \{0, 1\}^* \longrightarrow \mathcal{G} \quad (6)$$

sends s to the maximal Einstein development of that data, or to a specified failure object if the constraints cannot be solved.

Postulate 4 (Gravitational Kolmogorov weight). Let U_G be a prefix-free universal Turing machine whose language contains a primitive solver for the Einstein constraints and evolution (a *GR-machine*). The gravitational Kolmogorov complexity of a sample is

$$K_G(s) = \min\{|p| : U_G(p) \text{ outputs a description of } \Gamma(s) \text{ to the resolution of } s\}. \quad (7)$$

The unnormalized weight of s is

$$w(s) = |\langle e_s | \Omega \rangle|^2 2^{-K_G(s)}, \quad (8)$$

where $|e_s\rangle$ is the skeleton vector (or POVM element) that produced s .

Postulate 5 (Merge). The physical universe is not a single sample. It is the merge of all samples against the normalized weights

$$W(s) = \frac{w(s)}{\sum_{s'} w(s')}. \quad (9)$$

The merge has a coherent form (path integral), a decoherent form (ensemble), and a typical form (what an observer inhabits). All three are defined by the same countable family $\{W(s), \Gamma(s)\}$.

The cycle is Figure 1. QFT provides $|\Omega\rangle$ and the integers. Sampling reads a string. GR writes a continuum. K_G scores the writing. The merge reads the scores and returns physics.

4 Sampling: from $\aleph_0$ to $\mathfrak{c}$

4.1 What is sampled

The vector $|\Omega\rangle$ is not itself a universe. It is a countable list of amplitudes. To obtain a universe one must extract a definite string.

Let $\{E_s\}_{s \in \{0, 1\}^*}$ be a finite-resolution POVM on $\mathcal{H}$ indexed by strings of length at most N , refining as $N \rightarrow \infty$ to a complete measurement in a skeleton basis. A sample of resolution N is a draw s with Born probability

$$\Pr(s | \Omega, N) = \text{tr}(E_s |\Omega\rangle\langle\Omega|). \quad (10)$$

In a rank-one skeleton measurement this is $|\langle e_s | \Omega \rangle|^2$. The resolution N is Planck-scale information capacity, not an observer's laboratory precision: it is the number of bits the integer skeleton offers before inflation.

The string s is Cauchy data. Concretely, one may take s to encode:

- a finite simplicial 3-complex, or a finite causal set, or a finite graph with holonomies (a spin network), together with
- matter field values on the vertices, to a finite number of bits, and
- a finite specification of lapse and shift, or a statement that they are to be solved for.

The theory does not prefer a discretization. Any countable encoding of 3-data is a choice of skeleton. Uniformly equivalent encodings change K by $O(1)$ and leave the merge invariant at leading exponential order.

4.2 Inflation is Cantor’s map

The inflation map Γ is the physical form of (2). A binary sequence is a real. A countable list of Cauchy data is a spacetime, once Einstein’s equation is allowed to run.

Definition 4.1 (Inflation). If s encodes constraint-satisfying data (γ, K, Φ) on a 3-manifold, $\Gamma(s)$ is a representative of the diffeomorphism class of the maximal globally hyperbolic development. If the constraints fail, $\Gamma(s)$ is the development of the nearest constraint-satisfying projection of the data (in a fixed Sobolev metric), and the failure is charged to K_G as extra program length; see (11). If the development is incomplete in a non-physical way (naked singularity, closed timelike curves), Γ returns that incomplete object and K_G pays for the pathology.

Two structural facts make this a map $\aleph_0 \rightarrow \mathfrak{c}$ rather than a trick.

First: the domain is countable. There are $\aleph_0$ finite strings. There are $\aleph_0$ programs. Sampling cannot emit an uncountable object. Whatever continuum structure a universe has, it must be *generated* from a countable seed.

Second: the Einstein Cauchy problem is a completion. It takes data on a set that can be chosen countable-dense (analytic data; or a finite net, in the finite-resolution case) and produces a metric at every point of a 4-manifold of cardinality $\mathfrak{c}$. This is the same logical shape as completing $\mathbb{Q}$ to $\mathbb{R}$, or taking a power set. The continuum of GR is not an independent infinity floating beside QFT. It is the Cantor image of QFT’s integers, under the map “solve Einstein.”

Proposition 4.2 (Cantor inflation). *The image $\Gamma(\{0,1\}^*)$ is a countable subset of $\mathcal{G}$. Its metric completion, in any topology in which Einstein developments of refining data converge, has cardinality at most $\mathfrak{c}$ and contains the classical spacetimes that finite-resolution observers can stably identify.*

The physical continuum is a completion of a countable set of inflations, just as $\mathbb{R}$ is a completion of $\mathbb{Q}$. Generic elements of $\mathcal{G}$ —metrics that are not Einstein developments of any finite string—have $K_G = \infty$ in the sense that no program of the GR-machine prints them to arbitrary resolution. They have Solomonoff weight zero. They are the non-measurable, non-computable geometries that the gravitational path integral should never have been summing over.

4.3 The quantum state of the universe

Postulate 1 leaves $|\Omega\rangle$ unspecified up to the demand that it live in $\mathcal{H}$. A theory of everything must pick one.

Definition 4.3 (Kolmogorov vacuum). $|\Omega\rangle$ is a diffeomorphism-invariant state of least gravitational Kolmogorov complexity among states that admit a Fock representation of the matter fields on some Einstein background in the support of W . Equivalently, $|\Omega\rangle$ is a shortest program that prints a Hartle–Hawking no-boundary state [29]: compact Euclidean 4-geometries with a single boundary, no other boundary.

The no-boundary proposal is already a short program. That is why it is a candidate. The tunneling proposal [48] is another short program, of comparable length. The merge does not require us to decide between them at the level of postulates: both are samples, both receive weights, and W concentrates on whichever is shorter once the Einstein solver is in the language. If they have comparable K_G , both survive in the decoherent merge as alternative cosmological branches.

This is the sense in which the theory does not put in a wave function of the universe by hand. It puts in a universal machine with Einstein’s equation, and $|\Omega\rangle$ is a shortest vector that machine can name.

5 Weight: GR/ k -complexity

The user of a theory of everything is entitled to one number per universe. That number is $w(s)$ in (8). It has two factors, and they are not independent.

5.1 The Born factor

$|\langle e_s | \Omega \rangle|^2$ is the quantum sampling measure. It is the probability that the integer skeleton, interrogated at finite resolution, returns s . Without this factor the theory would be a classical algorithmic cosmology in the style of Schmidhuber [44]: possible, but silent about interference, identical particles, and the observed linearity of quantum amplitudes. With this factor, sampling is a measurement of $|\Omega\rangle$. The many worlds of Everett [20] are the samples. The present theory adds a second weight on those worlds, and a merge.

5.2 The gravitational Kolmogorov factor

$2^{-K_G(s)}$ is Occam’s razor computed on a GR-machine. Three comments.

Why a GR-machine, not a bare Turing machine. On a bare machine, a spacetime is a huge array of metric components, and $K(s)$ is essentially the raw bit-length of that array unless a compression happens to exist. Einstein’s equation *is* that compression, but only if the machine is allowed to use it. Putting the Einstein solver in the language of U_G is the statement that GR is fundamental, not emergent from a GR-blind program search. (If GR were itself a short program on a bare machine, the invariance theorem would absorb it into an $O(1)$ and the two choices would agree. If it is not, the GR-machine is a different theory, and it is the theory we mean.)

The split of K_G . Write, schematically,

$$K_G(s) = K(s_{\text{data}} \mid \text{Einstein}) + \frac{I_{\text{off}}[\Gamma(s)]}{\hbar \ln 2} + O(1). \quad (11)$$

The first term is the program length of the Cauchy data given that Einstein evolution is free. The second term charges constraint violation and off-shellness: I_{off} is a non-negative functional that vanishes on Einstein solutions and grows with $\int |G_{\mu\nu} - 8\pi T_{\mu\nu}|^2$. On shell,

$I_{\text{off}} = 0$ and K_G reduces to the complexity of the data plus the constant L_{Einstein} . Off shell, the extra bits kill the weight exponentially.

Action as complexity. The Euclidean Einstein–Hilbert action

$$I_E[g] = -\frac{1}{16\pi G} \int_M R \sqrt{g} d^4x - \frac{1}{8\pi G} \int_{\partial M} K \sqrt{h} d^3x \quad (12)$$

is the continuum shadow of K_G . Hartle and Hawking assign a universe the amplitude $e^{-I_E/\hbar}$ [29]. Dzhunushaliev replaces the action in the path integral by $K[g]$ [17]. The present theory says these are the same assignment once units are fixed by

$$\frac{I_E[\Gamma(s)]}{\hbar} \sim K_G(s) \ln 2 \quad \text{on the dominant saddles.} \quad (13)$$

This is a matching condition, not an identity for every metric. It is the statement that the semiclassical gravitational measure and the Solomonoff measure agree where both are reliable: on Einsteinian saddles of low complexity. Away from those saddles Solomonoff is the definition, and the continuum action is an approximation.

Proposition 5.1 (Einsteinian dominance). *Let s_E encode constraint-satisfying data of bit-length n , and let s_{rand} encode a generic metric array of the same resolution, not an Einstein development of any shorter string. Then*

$$K_G(s_E) \leq n + L_{\text{Einstein}} + O(1), \quad K_G(s_{\text{rand}}) \geq n - O(\log n), \quad (14)$$

and the weight ratio satisfies

$$\frac{w(s_{\text{rand}})}{w(s_E)} \leq 2^{-L_{\text{Einstein}} + O(\log n)} \cdot \frac{|\langle e_{s_{\text{rand}}} | \Omega \rangle|^2}{|\langle e_{s_E} | \Omega \rangle|^2}. \quad (15)$$

If $|\Omega\rangle$ is not conspiratorially supported on random geometries, Einsteinian samples dominate the merge by a factor exponential in the program length of Einstein’s equation.

That is why classical GR exists. It is not because the continuum was there first. It is because the shortest programs that print a 4-geometry print Einstein solutions.

5.3 Normalization

The denominator of (9) is a countable sum of non-negative terms. It may diverge. If it does, one passes to a sequence of finite-resolution cutoffs N (strings of length $\leq N$) and takes Banach limits of the finite- N weights, or equivalently uses the Solomonoff semimeasure without forcing $\sum W = 1$. This is standard [36]. The gravitational path integral has always had a measure problem [25]. Here the domain of summation is $\aleph_0$, not $\mathfrak{c}$, so the only remaining issue is the possible divergence of a series of positive terms, which is an ordinary analytic problem, not a foundational one.

6 Merge: from $\mathfrak{c}$ back to $\aleph_0$

Sampling produces a cloud of continuum universes, each tagged with a weight $W(s)$. Physics is what one does with the cloud. There are three consistent operations, and they are the three regimes of the theory.

6.1 Coherent merge: the k -integral

The coherent merge is the replacement for the gravitational path integral:

$$Z[J] = \sum_{s \in \{0,1\}^*} |\langle e_s | \Omega \rangle|^2 2^{-K_G(s)} \exp\left(\frac{i}{\hbar} S_{\text{matter}}[\Gamma(s); J]\right). \quad (16)$$

It is a countable sum. There is no continuum measure $\mathcal{D}g$ over metrics, no Faddeev–Popov determinant waiting on a non-existent Lebesgue measure on $\mathcal{G}$, no conformal-factor problem of the Euclidean integral as a *definition* (it may reappear as a property of the dominant samples). The continuum is inside $\Gamma(s)$, not in the domain of summation.

When K_G is matched to I_E as in (13) and the Born factor is slowly varying, (16) reduces to a skeleton-regularized Hartle–Hawking integral. The present definition is more general: it assigns weight to off-shell samples through K_G , and it knows about the quantum state $|\Omega\rangle$.

The effective amplitude for a coarse-grained 3-geometry h is

$$\Psi[h] = \sum_{s: \Gamma(s)|_{\Sigma} \simeq h} \sqrt{w(s)} e^{iS[\Gamma(s)]/\hbar}. \quad (17)$$

This Ψ is the wave function of the universe as an output, not an input. It is annihilated by the constraints to the extent that $\mathbf{U}_G$ treats diffeomorphisms as free (they do not add to K_G), which is the path-integral derivation of Wheeler–DeWitt [28] rewritten in Kolmogorov language.

6.2 Decoherent merge: the ensemble

Samples that differ by a macroscopic geometry do not interfere. The decoherence functional of Gell-Mann and Hartle [23] is, in this theory, a statement about K_G :

$$D(s, s') = \sqrt{W(s)W(s')} \exp(-K(s : s')), \quad (18)$$

where $K(s : s')$ is the mutual algorithmic information. When two inflations $\Gamma(s)$ and $\Gamma(s')$ have macroscopic metric differences, a short program cannot convert one into the other, $K(s : s')$ is large, and D is diagonal. The decoherent merge is then the mixed state

$$\rho_{\star} = \sum_s W(s) |U_s\rangle \langle U_s|, \quad |U_s\rangle = \text{a code of } \Gamma(s). \quad (19)$$

Classical spacetime is a diagonal ensemble of Einsteinian inflations. This is also the sense in which the theory is a many-worlds theory with a preferred measure: the worlds are the samples, the measure is W , and the preferred basis is the one that diagonalizes D , i.e. the low-mutual-information inflations.

6.3 Typical merge: the inhabited universe

An observer does not see $\rho_{\star}$. An observer inhabits a single inflation $\Gamma(s_{\star})$, where $s_{\star}$ is W -typical:

$$s_{\star} \in \{s : -\log_2 W(s) \leq K_G(s) + K(\Omega) + O(1)\}. \quad (20)$$

Typicality in the Solomonoff measure is the precise form of “the simplest universe consistent with the quantum state.” The inhabited geometry $\bar{g} = \Gamma(s_{\star})$ is the background of effective field theory. QFT, in the ordinary textbook sense, is the residual integer dynamics of $|\Omega\rangle$ in the Fock representation on $(\bar{M}, \bar{g})$.

Operationally, every prediction is a weighted expectation

$$\langle \mathcal{O} \rangle = \sum_s W(s) \mathcal{O}[\Gamma(s)]. \quad (21)$$

When W concentrates on a tight family of diffeomorphic Einstein solutions, (21) is the classical value of $\mathcal{O}$ on that solution, plus QFT fluctuations from the residual state. When W is spread (near a topology change, near a black-hole singularity, in the Planck regime), (21) is a genuine ensemble average and classical geometry fails.

6.4 The merge returns a quantum state

The output of the merge is again an object in a separable Hilbert space: $\rho_\star$ in (19), or Ψ in (17). The cycle in Figure 1 closes. This is required for consistency. If the merge produced a naked continuum object with no countable description, the theory would have used $\mathfrak{c}$ as an output while forbidding it as an input. Instead:

$$\aleph_0 \xrightarrow{\Gamma} \mathfrak{c} \xrightarrow{W} \aleph_0. \quad (22)$$

The continuum is an intermediate, generated and then integrated out. That is what a well-defined gravitational path integral always should have been.

7 Recovery of general relativity

Classical GR is the typical merge of low- K_G samples.

1. **Einstein’s equation.** Proposition 5.1: the GR-machine prints Einstein developments far more cheaply than generic metrics, so W concentrates on $\text{Ein}(\mathcal{G}) \subset \mathcal{G}$. Off-shell samples survive only at Planckian resolution, where $I_{\text{off}}/\hbar$ is not large.
2. **Diffeomorphism invariance.** A diffeomorphism is a change of coordinates. On the GR-machine it is a free subroutine: applying a diffeomorphism to a named metric does not increase K_G by more than $O(1)$ plus the complexity of the diffeomorphism itself, and the diffeomorphisms that are namable by short programs are the ones physics already treats as gauge. Two samples that inflate to diffeomorphic spacetimes are identified in $\mathcal{G}$ (Postulate 2), so they are the same universe.
3. **Local well-posedness and causality.** Inflation *is* the Cauchy problem. Globally hyperbolic developments are exactly the objects Γ produces on shell. Closed timelike curves require extra data (identifications) that increase K_G ; they are not forbidden by fiat, they are expensive.
4. **The Einstein–Hilbert action as Hamilton–Jacobi function.** On the dominant samples, (13) holds, so the phase of the coherent merge is the Einstein–Hilbert phase, and the WKB limit of $\Psi[h]$ is classical GR with that action. Lovelock uniqueness [37] says that in four dimensions the only second-order diffeomorphism-invariant tensor built from the metric is Einstein’s. A GR-machine whose Einstein solver was replaced by a higher-order PDE would have a longer program (more jets to specify) or would lose well-posedness (and then pay I_{off}). Four-dimensional Einstein gravity is the shortest well-posed compressor of 3-data into 4-geometry.

5. **Matter coupling.** The matter content of $\Gamma(s)$ is whatever field data s carried, evolved by the coupled Einstein–matter system. The shortest matter programs compatible with the residual quantum state are the ones that appear in the typical universe. This is not a derivation of the Standard Model. It is a statement of what would count as a derivation: a demonstration that the Standard Model Lagrangian is a shortest matter plugin for $\mathcal{U}_G$ consistent with $|\Omega\rangle$ and with the observed low-energy sample.

8 Recovery of quantum field theory

QFT is the integer theory on a merged background.

Fix a typical inhabited geometry $(\bar{M}, \bar{g}) = \Gamma(s_*)$. The residual quantum state is the condition of $|\Omega\rangle$ on the skeleton, given s_* . In a neighbourhood of s_* in which inflation produces geometries in which a Fock representation exists (Hadamard states on globally hyperbolic backgrounds [49, 33]), this residual state *is* a QFT state of the matter fields, and of gravitons as linearized excitations of $\bar{g}$.

1. **Particles are integers.** Occupation numbers, species indices, momenta in a box: $\aleph_0$. There is no continuum of particle types because a continuum of types would be a $\mathfrak{c}$ -sized label set, which sampling cannot emit. A “continuous spin” or a continuous family of species can appear only as a classical limit of many discrete samples.
2. **Fields are continuum completions.** A quantum field operator is indexed by test functions on $\bar{M}$, and $\bar{M}$ has cardinality $\mathfrak{c}$. This is inflation again: the field as a distribution is the GR-plus-matter completion of countable mode data. The notorious infinities of QFT are $\aleph_0$ -sums (mode sums, loop momenta on a lattice going to infinity) pretending to be $\mathfrak{c}$ -integrals without having passed through the merge. The merge caps them.
3. **Ultraviolet divergences and holography.** A local QFT Hilbert space is infinite-dimensional because a region of continuum spacetime supports $\aleph_0$ modes. Gravity does not permit this. The Bekenstein bound and its covariant forms [4, 8], and the independent argument that a local region in quantum gravity has finite-dimensional Hilbert space [2], say that a region of boundary area A supports at most $A/4\ell_P^2$ independent bits. In the present theory that statement is the assertion

$$K_G(s|_R) \leq \frac{A(\partial R)}{4\ell_P^2} + O(\log A). \quad (23)$$

A sample that tries to specify more independent data in R than the area allows either inflates to a geometry in which R has already collapsed to a black hole (and the extra data are behind a horizon, charged to a different region), or it is not a shortest program and is exponentially downweighted. The continuum interior of R is a Γ -completion of a countable boundary code of length $\sim A/4\ell_P^2$. That is holography [47, 46, 38] as Cantor inflation with a Kraft bound.

4. **The Born rule.** Laboratory Born frequencies are the W -weights of samples that differ only in a small subsystem, with the environment and the geometry held fixed. For those samples K_G is essentially constant (the extra bits specifying which detector clicked are $O(1)$ compared with the complexity of the apparatus), so $W(s) \propto |\psi_s|^2$. Corrections are considered in Section 13.

5. **QFT in curved spacetime.** The typical background $\bar{g}$ need not be Minkowski. The residual state on $\bar{g}$ is ordinary QFT in curved spacetime [6, 49]. Hawking radiation is the residual integer dynamics on a collapsing typical sample [32]. Inflationary perturbations are the residual dynamics on a typical inflating sample [41].

9 Black holes

A Schwarzschild sample of mass M is a very short string: one real number $r_g = 2GM$, plus Einstein’s equation, plus a statement of vacuum and spherical symmetry [17]. Spherical symmetry is itself a short program (“all points at radius r are equivalent”). Hence Schwarzschild, Kerr, and FLRW are not generic points of $\mathcal{G}$. They are the Kolmogorov elite. That is why they are the first solutions anyone writes down, and why they dominate the merge among macroscopic isolated objects.

The horizon is the place where the integer skeleton of the exterior ends. A sample of the exterior of a black hole of area A cannot cheaply specify independent interior data beyond $A/4\ell_P^2$ bits: (23) applied to the exterior region. The gravitational Kolmogorov complexity of the hole, as an object in the merge, is therefore

$$K_G(\text{BH of area } A) = \frac{A}{4\ell_P^2} + L_{\text{Einstein}} + K(\text{charges, angular momenta}) + O(\log A). \quad (24)$$

The first term is Bekenstein–Hawking entropy [3, 32, 24]. In this theory it is not a thermodynamic analogy. It is the program length of the shortest GR-machine description of the hole.

The decoherent merge of samples that agree on the exterior and differ on the interior by up to $e^{A/4\ell_P^2}$ bits is a mixed state of von Neumann entropy $A/4\ell_P^2$. That mixed state is the Hawking radiation thermal state, viewed from the outside. Information is not lost in the coherent merge: the coherent sum (16) still contains the interior samples. It is lost, operationally, in the decoherent merge, because $K(s_{\text{ext}} : s_{\text{int}})$ is large and the off-diagonal terms in D vanish. Unitary evaporation is the statement that at late times a new family of samples—the radiation—becomes a shortest description of the same universe, with K_G matching (24). Whether that matching is exact (no firewall, no remnant) is a question about the GR-machine’s shortest programs for evaporating geometries, not a question that has to be settled before the theory can be stated.

10 Cosmology and the arrow of time

10.1 The typical cosmos

A typical sample under W has:

- short Cauchy data (low $K(s_{\text{data}})$),
- on-shell Einstein inflation ($I_{\text{off}} = 0$),
- a residual quantum state close to the Kolmogorov vacuum.

Short, constraint-satisfying, homogeneous data on a 3-sphere or a torus, plus a short inflaton potential, are extremely cheap programs that inflate to large, smooth, approximately FLRW universes. This is why the observed cosmos is large, smooth, and approximately isotropic. It is not because a continuum of initial conditions was fine-tuned. It is because the alternatives are longer programs.

Inflation is preferred as a compressor: a few e-folds of a short potential convert Planckian data into a vast Einsteinian region, which would otherwise have to be specified bit by bit. Too much inflation, of a kind that requires a long fine-tuned potential or an enormous number of named e-folds, costs K_G . The measure W therefore favours a moderate inflating family, in the same qualitative direction as the Hartle–Hawking measure [30] and opposite to volume-weighting that rewards $10^{10^{10}}$ e-folds. A quantitative N_e is a computation of K_G for concrete inflaton plugins; it is not settled by the postulates.

10.2 The cosmological constant

A generic vacuum energy is an incompressible real: specifying Λ to N bits costs N bits. A vacuum energy that is a short program—0, or a topological invariant, or a ratio of two short geometric quantities—is cheap. The typical merge therefore does not produce a Planck-scale cosmological constant as a random QFT zero-point sum. That sum is an $\aleph_0$ -divergence that the merge never performs (Section 8). What it produces is a Λ with a short description, possibly zero, possibly the residual of a short compactification, possibly the smallest nonzero value namable by the matter plugin. The observed value $\Lambda \sim 10^{-122}$ in Planck units is a 122-bit number and is *not* short as a raw integer. The theory’s claim is that the shortest program for the observed Λ is not “print this 122-bit integer” but a short geometric or vacuum-structure program whose output happens to be that integer. If no such program exists, the observed Λ is a genuine cost, and the typicality of our sample has to be paid for elsewhere (anthropic conditioning on the residual state, which the theory permits but does not advertise).

10.3 The arrow of time

Kolmogorov complexity of a typical inhabited universe, as a function of Cauchy time, is low on the initial slice—that is what “typical under W ” means—and high later, because Einstein evolution of even short data produces geometrically rich 4-regions whose uncompressed description is long. The algorithmic entropy of the slice [51] increases. That is the arrow of time. It does not require a special postulate about past hypotheses beyond the statement that W itself is the past hypothesis: the measure is supported on short data.

Boltzmann brains are long incompressible strings that fluctuate into existence in a typical late-time geometry. Their K_G is of order their own bit-length, so W suppresses them exponentially relative to ordinary cosmological observers, who are cheap by-products of a short inflating program [12]. This is the standard algorithmic resolution, now sitting on the same weight that defines the theory.

11 The continuum hypothesis, physically

Sampling emits elements of a countable set. Inflation produces objects of cardinality $\mathfrak{c}$. The merge sums over a countable index. No construction in the theory produces a set of cardinality κ with $\aleph_0 < \kappa < \mathfrak{c}$.

Proposition 11.1 (Physical CH). *Every set that the k -cycle constructs from $|\Omega\rangle$ by sampling, inflation, weighting, and merging is either finite, of cardinality $\aleph_0$, or of cardinality $\mathfrak{c}$ (or, if one takes power sets of geometries, of cardinality $2^{\mathfrak{c}}$, which the cycle never samples). In particular the theory has no physical use for a cardinal strictly between $\aleph_0$ and $\mathfrak{c}$.*

This is not a proof of CH in ZFC, which is impossible [26, 16]. It is a statement that physics, if it is this theory, is a model of set theory in which the only infinite cardinals that

occur as types of physical objects are $\aleph_0$ and $\mathfrak{c}$. Intermediate cardinals, if they exist, are empirically idle. That is the strongest physical meaning CH can have.

It is also a diagnosis of a class of failed unifications. A theory that quantizes GR by putting a Hilbert space of dimension $\mathfrak{c}$ (a non-separable kinematical space of “all metrics”) on top of the continuum, without a sampling map back to $\aleph_0$, has used a cardinal that QFT does not know how to measure. Loop quantum gravity’s kinematical Hilbert space is non-separable in the standard construction and becomes separable once diffeomorphism moduli are treated as unphysical [21]; that move is, in the present language, a refusal to treat a $\mathfrak{c}$ -sized label set as quantum kinematics. The refusal is correct.

12 What the theory is not

It is not a derivation of the Standard Model spectrum, gauge group, or three generations. Those are matter plugins to U_G . The theory says how a plugin is weighted, not which plugin is shortest.

It is not a claim that Kolmogorov complexity is computable, or that observers compute K_G in order to exist. W is a defining measure, as the Euclidean action is a defining measure in Hartle–Hawking. Predictions that require the exact value of K are not available; predictions that require only crude bounds (Einsteinian dominance, holographic caps, shortness of Schwarzschild, suppression of Boltzmann brains) are.

It is not hypercomputation. The GR-machine is a Turing machine with an Einstein primitive. If that primitive is itself Turing-computable (numerical GR to finite resolution, which is all sampling ever needs), U_G is an ordinary UTM and K_G differs from K by $O(1)$. The continuum is never computed pointwise. It is named by a program that *refers* to an Einstein development.

It is not, despite the countable sums, a causal-set theory [7], a CDT theory [1], or a Wolfram model [50], although each of those is a legitimate choice of integer skeleton. The ontology is the pair $(\mathcal{H}, \mathcal{G})$ with maps Γ and W , not a particular discretization.

It is not a solution of Gödel–Chaitin incompleteness [14, 22]. K_G is uncomputable, $|\Omega\rangle$ is named but not constructed, and there will be physical questions whose answers are not theorems of any recursive axiomatization of the GR-machine. The theory is complete as an ontology and a measure. It is incomplete as a decision procedure. That is the correct amount of completeness for a theory that takes Kolmogorov complexity as fundamental.

13 Predictions and tests

A theory of everything earns the name by recovering the two existing theories and then saying something they do not. The recoveries are Sections 7–10. The extras are these.

1. The gravitational path integral is a countable sum. Any continuum measure $\mathcal{D}g$ is an approximation to (16), valid on Einsteinian saddles and nowhere a definition. Practical consequence: lattice, spin-foam, and causal-dynamical-triangulation sums are closer to the definition than continuum Euclidean integrals are. Their continuum limits, when they exist, are inflations, not domains of integration.

2. A complexity correction to Born frequencies. For two laboratory outcomes a, b of a measurement on a small system,

$$\frac{W(a)}{W(b)} = \frac{|\psi_a|^2}{|\psi_b|^2} 2^{-(K_G(a) - K_G(b))}. \quad (25)$$

In a laboratory whose environment has already contributed $\sim 10^{23}$ bits to K_G , the difference $K_G(a) - K_G(b)$ is $O(1)$ and the correction is unobservable. In a fully specified, isolated, few-qubit device whose total K_G is dominated by the register, a systematic bias toward the lower-complexity outcome is in principle present. This is the most direct empirical signature the theory has, and the easiest to overclaim. It should be looked for only in experiments where the Kolmogorov complexity of the two branches can be bounded independently of the Born weights (for example, one branch is a short computational history and the other is an incompressible dump).

3. Holographic, not field-theoretic, vacuum energy. Zero-point sums are not performed. The residual Λ is a short program's output, and the UV cutoff of matter loops in a region is the area bound (23), not a cutoff in momenta of a continuum of modes. This is already the standard holographic expectation [47, 2]; here it is a theorem of the merge.

4. Moderate inflation, simple potentials. The measure W rewards inflaton plugins that are short programs producing enough e-folds to solve the horizon problem, and penalizes both no inflation (long data on a large slice) and baroque landscapes named pointwise. A single-field monomial or exponential of low degree is favoured over a random 10^{500} -vacuum list unless the list itself has a short generator. This is a selection principle on the landscape, not a unique potential.

5. Macroscopic solutions are Kolmogorov-elite. The observed catalogue—Kerr, FLRW, gravitational waves as linearized Einstein radiation, no naked singularities—is the catalogue of short GR-machine programs. A stable, macroscopic, non-Einstein, non-diffeomorphic-to-Einstein geometry would falsify Einsteinian dominance (Proposition 5.1) and so falsify the theory.

6. No physical intermediate cardinals, no non-separable kinematics. A compelling empirical role for a non-separable Hilbert space, or for a physical observable whose spectrum has cardinality strictly between $\aleph_0$ and $\mathfrak{c}$, would falsify Postulates 1–3. (Continuous spectra of ordinary operators are not a counterexample: they are $\mathfrak{c}$ -sized, and they arise as inflations of countable spectral data, as in the spectral theorem for $L^2(\mathbb{R})$.)

7. Black-hole entropy is program length. Any microscopic counting of black-hole states that does not, at leading order, reproduce (24) as a Kolmogorov bound is counting something other than the weight the merge uses. Agreement of string, loop, and Euclidean calculations with $A/4\ell_P^2$ is, on this view, agreement that those calculations are enumerating shortest GR-machine descriptions.

14 Related work

The theory sits at a crossing that several programs have approached from one side.

Algorithmic cosmologies. Solomonoff's universal prior [45], Levin's semimeasure [35], and Schmidhuber's algorithmic theories of everything [44] put 2^{-K} on universe-histories. They do not put QFT and GR at two Cantorian infinities, they do not inflate samples with the Einstein Cauchy problem, and they do not merge. Müller's algorithmic idealism [39, 40] starts from observer states and Solomonoff prediction; the present theory starts from $|\Omega\rangle$ and a GR-machine, and treats observers as typical samples.

Complexity in quantum gravity. Dzhunushaliev’s identification of gravitational probability with $e^{-K[g]}$, his rewriting of the path integral, and his Schwarzschild estimate [17, 18] are the nearest ancestors of Postulate 4. The present theory adds the cardinal split, the sampling of a separable $|\Omega\rangle$, the three merges, and the identification of holography with a Kraft bound on K_G . Nielsen (circuit) complexity and Krylov complexity in cosmology [5, 42] are different quantities: they measure depth of a unitary, not program length of a universe. They may bound K_G in special states; they do not replace it.

Quantum cosmology. The Hartle–Hawking no-boundary state [29], Vilenkin’s tunneling state [48], and decoherent histories [23, 30] are the coherent and decoherent merges in a different language, with I_E in place of K_G . The present theory claims that I_E is the continuum shadow of K_G on saddles, and that the sum is over programs, not over a continuum of metrics.

Holography and finite-dimensional gravity. ’t Hooft, Susskind, Maldacena [47, 46, 38]; Bousso’s bound [8]; Bao, Carroll, and Singh’s locally finite-dimensional Hilbert space [2]. Equation (23) is that circle of ideas stated as a bound on gravitational Kolmogorov complexity of a sample.

Discrete quantum gravity. Causal sets [7], CDT [1], loop quantum gravity [43, 21], and spin foams are candidate integer skeletons and candidate inflation maps. The present theory is agnostic among them so long as Γ is an Einstein completion and $\mathcal{H}$ is separable.

Cantorian spacetime. El Naschie’s E -infinity Cantorian spacetime [19] uses transfinite Cantor sets as a model of spacetime and is a different project: it replaces the manifold, rather than treating the manifold as an inflation of an integer quantum skeleton.

15 A discrete analog

The k -cycle can be run in miniature, with no claim that the miniature is gravity.

Let $\mathcal{H} = \mathbb{C}^{2^n}$ with the computational basis $\{|s\rangle : s \in \{0, 1\}^n\}$ as integer skeleton. Let $|\Omega\rangle$ be a state of low circuit complexity (a short-program state). Sampling is a computational-basis measurement, producing a bitstring s of length n . Inflation Γ is, in the analog, the discrete Poisson problem—the nearest cousin of a constraint equation—or simply the identification of s with a classical spin configuration on a cycle of length n . Gravitational Kolmogorov complexity is approximated by

$$\widehat{K}_G(s) = \ell_{\text{gz}}(s) + \beta E_{\text{smooth}}(s), \quad (26)$$

where ℓ_{gz} is the compressed length of s (a computable upper bound on K) and $E_{\text{smooth}} = \sum_i (s_{i+1} - s_i)^2$ is a discrete curvature penalty, the analog of I_{off} . Weights $W(s) \propto |\langle s|\Omega\rangle|^2 2^{-\widehat{K}_G(s)}$ are formed, and the merge is the weighted barycentre $\bar{s} = \sum_s W(s)s$ in $[0, 1]^n$, or the typical sample.

Three things happen, and they are the three things the continuum theory claims. Low-complexity, smooth strings dominate. The merge is a finite sum. The output is again a vector in a separable space (here finite-dimensional). A companion program in this repository runs that analog.

The analog is also a warning. Gzip is not K , a cycle graph is not Einstein, and $n < \infty$ erases the cardinal split. What survives is the shape of the cycle: sample from a quantum state, score by compression plus a geometric penalty, merge.

16 Open problems

1. **The Einstein primitive.** Specify the GR-machine in enough detail that L_{Einstein} is a definite (if enormous) integer, and prove a precise version of (13) for Euclidean saddles.
2. **Uniqueness of Γ .** Different discretizations of Cauchy data give different maps Γ . Prove that their merges agree on all diffeomorphism-invariant coarse observables in the $\hbar \rightarrow 0$, $N \rightarrow \infty$ limit.
3. **The Kolmogorov vacuum.** Construct, or at least uniquely characterize, $|\Omega\rangle$ as a shortest diffeomorphism-invariant state. Decide the no-boundary versus tunneling contest as a K_G comparison.
4. **Standard Model as plugin.** State a precise minimization problem whose solution would be the Standard Model: a shortest matter Lagrangian, in a fixed encoding of field theories, compatible with a given low-energy sample.
5. **Evaporation.** Compute, in a controlled analog or a minisuperspace model, whether K_G of an evaporating hole plus radiation remains equal to $A_{\text{initial}}/4\ell_P^2$ along the history. This is the information problem as a problem in gravitational Kolmogorov complexity.
6. **Complexity corrections.** Bound $K_G(a) - K_G(b)$ in a concrete few-qubit interferometer and say whether (25) is experimentally accessible.
7. **Dimensionality.** Show that $3 + 1$ is the shortest well-posed Einstein compressor, in the sense of Dzhunushaliev's complexity-driven dimensional reduction [17, 18] made into a theorem about K_G .

17 The theory in one page

Cantor split physics into two infinities before physics knew it needed them. The integers index quantum states. The reals index spacetime. The reals are the power set of the integers, so a universe is a subset of the quantum skeleton, completed by Einstein's equation.

Sample a finite string from $|\Omega\rangle$. Inflate it with GR to a continuum geometry. Weight it by 2^{-K_G} times its Born factor, where K_G is Kolmogorov complexity on a machine that already knows Einstein. Merge the weighted cloud. The merge is a countable sum. Classical spacetime is the typical term. Quantum field theory is the integer residue. Holography is Kraft's inequality on a region of finite area. Black-hole entropy is program length. The arrow of time is the unfolding of a short program. The gravitational path integral was never an integral over the continuum. It was always this sum.

That is a theory of everything in the only sense the phrase can bear. It names the ontology, the measure, and the maps between the two infinities the twentieth century actually used. It recovers GR as Einsteinian dominance of the merge, and QFT as residual integer dynamics. It leaves the Standard Model as a shortest-plugin problem, K_G as uncomputable, and $|\Omega\rangle$ as a shortest vector rather than a constructed one. Those are debts. They are debts of the right shape.

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